CSE370 Database Systems: Lab Assignment 3

Objective	Practice advanced SQL operations and JOIN using multiple tables in MySQL.		
Due Date:	1 Week, submit before next class	**Number of Questions:**	10

Scenario:

You are working as a data analyst for a library network that operates multiple branches across different cities. The library maintains records of its members, the books held at each branch, books donated by members, and study room bookings. Your task is to write SQL queries using joins to extract information from the interconnected data across multiple tables.

Part 1: Understanding the Data

Tables: member, branch, book, donor, room_booking, booking_member

Find the table schema and sample data here: CSE370: Lab Asmt 3 Data

Follow the instructions given in the document to create the tables and insert data into them.

```
Library Database Schema:

Member (member_id, member_name, member_street, member_city) -- PK: member_id

Branch (branch_name, branch_city, budget) -- PK: branch_name

Book (book_id, title, branch_name, price) -- PK: book_id
-- FK: branch_name references Branch(branch_name)

Room_Booking (booking_id, branch_name, fee) -- PK: booking_id
-- FK: branch_name references Branch(branch_name)

Donor (member_id, book_id) -- PK: member_id, book_id
-- FK: member_id references Member(member_id)
-- FK: book_id references Book(book_id)

Booking_Member (member_id, booking_id) -- PK: member_id, booking_id
-- FK: member_id references Member(member_id)
-- FK: booking_id references Room_Booking(booking_id)
```

Part 2: Questions

IMPORTANT: Provide one single query per question! Your query should not be hard coded, it should provide the logic mentioned in the question.

Write SQL queries to complete the following tasks: [10 * 1 = 10]

Q1. Find the name and booking ID of all members having a room booking at the Central branch.

Q2. Find the name of members who have made a room booking.

Q3. Find the name of members who have not made a room booking.

Q4. Find all the possible pairs of members who are from the same city. Show in the format Member1, Member2, City.

Q5. If the library charges a 5% maintenance cost on the price of every book, show the total maintenance cost across each branch. Print Branch_name, Total_Maintenance (rounded to 2 decimal places).

Q6. Among the books donated by members, find the book ID with the highest price for each member city in the database. Show member_city, book_id, price.

Q7. Show the booking ID, booking fee, and name of members with the top 5 highest booking fees. The data should be sorted by increasing fee, then decreasing booking ID in case of the same fee.
[Hint for top 5: Check the "limit" keyword in MySQL]

Q8. Find the names of members who have donated a book held at the Lakeview branch and also have a room booking at the Lakeview branch.

Q9. Find the total booking fee of all members having at least 2 room bookings. Show in the format member_name, total_fee.

Q10. Find the member_id and sum of book prices of all members who have donated more than one book. Display member_id and total_price.

Submission Instructions

Make a copy of the `submission template` in your own Google Drive.

For each task in Part 2:

1. Write the MySQL command in your submission document.
2. Execute the command in XAMPP shell (MySQL command line), take a screenshot showing the command and output, then paste it below the command in your document. **[Your StudentID with your database name, must be visible in each screenshot, ELSE YOU WILL RECEIVE 0 FOR THIS ASSIGNMENT]**
3. After completing the 10 tasks, download your document as PDF.
4. Rename using format: `Lab_Sec##_ID_Name_CSE370_Assignment3.pdf`

   Example: `Lab_Sec12_12345678_PeterParker_CSE370_Assignment3.pdf`
5. Submit via the Google Form here: `https://forms.gle/sJENgNYraMQgd5DD7`